

EDUCATION	UNIVERSITY OF ORLÉANS <i>Ph.D., Computer Science and Signal Processing</i> • Dissertation: <i>Time Series Forecasting: From Econometrics to Deep Learning</i>	ORLÉANS, FRANCE 10/2019 – 09/2023
	UNIVERSITY OF ORLÉANS <i>Diplôme d'Ingénieur¹, Computer Engineering, Polytech Orléans</i> <i>M.Sc., Computer Science</i>	ORLÉANS, FRANCE 09/2015 – 09/2018 09/2017 – 09/2018
	BEIJING INSTITUTE OF TECHNOLOGY <i>B.Eng., Electrical & Electronics Engineering</i>	BEIJING, CHINA 09/2012 – 06/2016
PROFESSIONAL EXPERIENCE	MEITUAN <i>CLC, Search and Recommendation Dept., Algorithm Engineer (L7)</i> • Led a team of 3, deeply involved in optimizing and upgrading algorithmic pricing strategies for Meituan's membership coupon packages, responsible for the overall business in lower-tier markets, and in charge of iterative upgrades to the cost control system.	BEIJING, CHINA 12/2023 – Present
	UNIVERSITY OF ORLÉANS <i>Lecturer, Polytech Orléans</i> • Introduction to Signal Processing. Signal and Linear Systems. Acquisition Systems. MATLAB. • Arduino & Embedded Systems.	ORLÉANS, FRANCE 01/2023 – 09/2023
	ATTILA GESTION <i>Data Scientist</i> • ATTLA is a building industry franchise, partnering with major industry groups. Responsible for segmenting customer groups of different sizes and fields to allocate appropriate resources and services. Also responsible for forecasting key indicators such as the number of customers, order volume, and revenue to support business decision-making and operations.	LYON, FRANCE 04/2018 – 12/2022
SELECTED PROJECTS	OPTIMIZATION OF MARKETING STRATEGY FOR COUPON PACKAGE • Coupon package is an important marketing tool, aiming to boost growth targets by leveraging users' loss aversion psychology. The strategy indirectly lowers the package price through different distribution of discount, promoting the conversion of users and thus driving growth. • In the early stage of store integration ² , operations put manually differentiated offers to user segments based on <i>Order-AOV</i> combinations. The algorithm adopted an "ML + OR" framework for uplift modeling of different treatment actions. The main challenges included: 1) a large action space; 2) significant budget fluctuations; 3) difficulty in modeling the <i>Actual Payment</i> . • The v1.0 strategy, limited by RCT data volume, chose to model <i>Order</i> as the target. To address large differences between online and offline costs, high-subsidy treatment mappings were designed to expand the cost decision space. Online, three A/B tests were conducted: 1) Lagrangian allocation at the user segment level, 2) Lagrangian allocation at the user level, and 3) a robust, heuristic-based 0-1 treatment strategy. Solution 2 was fully rolled out, increasing CLC orders by +0.96% and actual payment by +0.42%. • The v2.0 strategy mainly optimized the GTV metric. As the GTV fluctuated greatly, it has a pronounced long-tail effect, and a low SNR. Therefore, the GTV target was decomposed as $GTV = Order \times AOV$, with separate modeling for each. A concave function transformation was applied to the AOV to enhance incremental signals in the main part of the probability density and reduce noise in the extremes, thus improving the SNR. In addition, the two-stage framework was replaced with an end-to-end DFL solution. This combined solution rolled out and increased CLC orders by +0.71% and actual payment by +0.32%. • The v2.2 strategy adopted the MMoE model, using multi-objective modeling to differentiate across business units benefiting from MMoE's multi-task learning capabilities. This solution was fully rolled out, increasing CLC orders by +0.35% and actual payment by +0.26%.	05/2024 – PRESENT

¹Industrial Ph.D. program contracted with ATTLA Gestion.

- Applied clustering methods to segment ATTILA's customers, and used econometric and DL approaches to forecast and evaluate metrics such as customer count, order volume, and revenue.
- Investigated the impact of decomposition-ensemble strategies on the performance of different forecasting methods, implemented an evaluation framework for DL models under various forecasting strategies, used to assess their performance on datasets with different features (*length, granularity, seasonality, stationarity*).
- Developed an automated forecasting toolkit automatically selecting optimal forecasting strategies (*ensemble learning, decomposition-ensemble strategies*) and models (*ARIMA, Theta, ES-RNN, Prophet, STLformer*) for time series data with different features/lengths.
- Encapsulated the above methods into a web application using Flask, Plotly, and sktime, where users only need to upload Excels in matching format to complete forecasting, with analysis displayed in multi-dimensional charts to support business decision-making and operations.

STLFORMER: FORECASTING WITH STL DECOMPOSITION AND RANK CORRELATION

03/2023

- Proposed STLformer, a Transformer-based time series forecasting model.
- STLformer progressively decomposes time series with STL decomposition and models seasonal and long-term trend patterns with an attention-based encoder and decoder, respectively.
- STLformer tests the ARCH effect and uses Spearman's ρ -based attention to model nonlinear dependencies.
- STLformer achieves forecasting in $\mathcal{O}(N \log N)$ and outperforms SOTA methods on multiple datasets, especially on nonlinear dependent series (e.g., financial series).

ON THE DECOMPOSITION-ENSEMBLE STRATEGY ON TSF ALGORITHMS

07/2021

- Assessed various decomposition methods, i.e., Classical, STL, and Prophet.
- Implemented econometric and ML models under the decomposition-ensemble strategy.
- Evaluated the performance of different models on the M-Competition dataset.
- This strategy can benefit traditional models, but its impact on ML models depends on the data distribution.

SELECTED PUBLICATIONS

1. **Z. Ouyang**, M. Jabloun, and P. Ravier, "STLformer: Exploit STL decomposition and Rank Correlation for Time Series Forecasting," in *Proc. EUSIPCO*, 2023.
2. **Z. Ouyang**, M. Jabloun, and P. Ravier, "Rankformer: Leverage Rank Correlation for Transformer-based Time Series Forecasting," in *Proc. IEEE SSP*, 2023.
3. G. Ouyang, K. Abed-Meraim, and **Z. Ouyang**, "Magnetic-Field-Based Indoor Positioning Using Temporal Convolutional Networks," *Sensors*, vol. 23, no. 3, p. 1514, 2023, (SCIE Q1, IF=3.9).
4. **Z. Ouyang**, P. Ravier, and M. Jabloun, "Are Deep Learning Models Practically Good as Promised? A Strategic Comparison of Deep Learning Models for Time Series Forecasting," in *Proc. EUSIPCO*, 2022.
5. **Z. Ouyang**, P. Ravier, and M. Jabloun, "STL Decomposition of Time Series Can Benefit Forecasting Done by Statistical Methods but Not by Machine Learning Ones," *Eng. Proc.*, vol. 5, no. 1, p. 42, 2021.

SKILLS**Programming:** Python, Java, Spark/Hive, SQL, $\text{\LaTeX}$ **Frameworks & Tools:** TensorFlow, PyTorch, scikit-learn, Unity3D, OpenCV, PowerBI, Linux, Git**Expertise:** Machine/Deep Learning, Time Series Analysis, Uplift Modeling, Causal Inference, Signal Processing, Nonlinear Optimization**Languages:** English (proficient), French (proficient), Mandarin (native)**AWARDS**

- Erasmus+ Consortium Polytech, Polytech Orléans 2017
- College Student Academic Scholarship, Beijing Institute of Technology 2012 – 2015
- National 3rd Prize, Chinese Exhibition of Calligraphy and Painting for Undergraduates 2013

OTHER EXPERIENCE

- Volunteer, Chinese New Year Festivity, *Orléans and Yangzhou Government* Feb. 2017
- Vice President, *Association of Calligraphy of Beijing Institute of Technology* 2013 – 2015

HOBBIES

Basketball, Reading, Chinese Calligraphy, Singing, Fitness, and Cooking.

²Integration of ToHome and ToStore businesses²"Diplôme d'Ingénieur" is a highly accredited elite diploma in France, equivalent to M.Eng. Only the top 10% of the students in the French Baccalaureate can apply for this education program in engineering schools. Students delve deep into engineering and science courses and receive management, economics, and social sciences education, ensuring they emerge as well-rounded professionals.